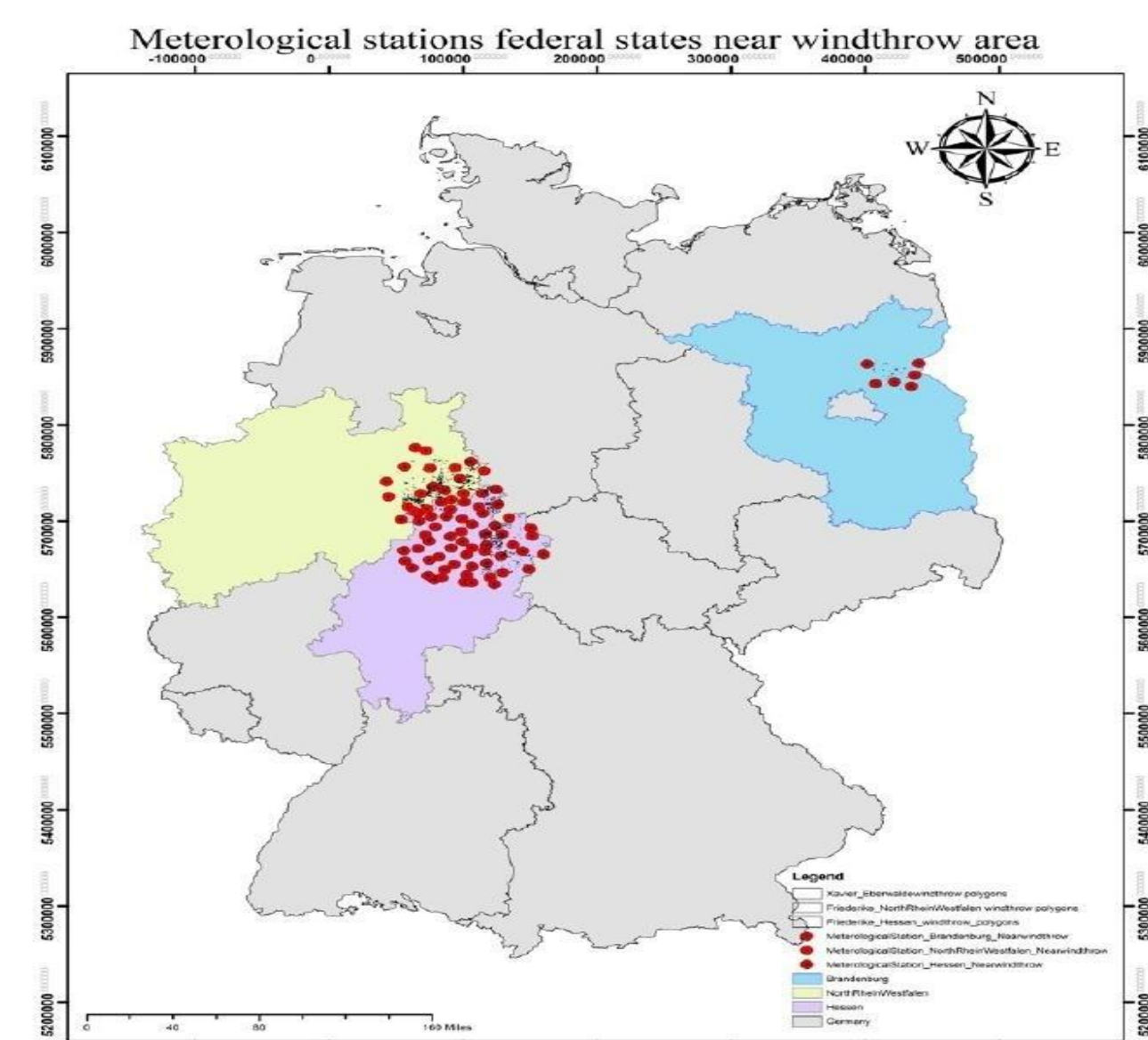


Windthrow Soil Moisture Pre-study

- Soil moisture values on the windthrow-damage polygons in the study region
- Spatial relationship between the windthrow polygons, depending on their nature
- Differences between the two satellite products' soil-moisture grid pixel values

Study area and storm event



Friederike
15–16 Jan 2018 — North Rhine-
Westphalia (NRW) & Hessen
Polygon sources: ESA (European Space
Agency) for both NRW and Hessen

Forest damage location
Based on several forest-damaged
(windthrow) polygons identified across
the three federal states.

Satellite 1 — Sentinel-1 (combined 1-A / 1-B product)

Satellite 1 — Sentinel-1 (combined 1-A / 1-B product)

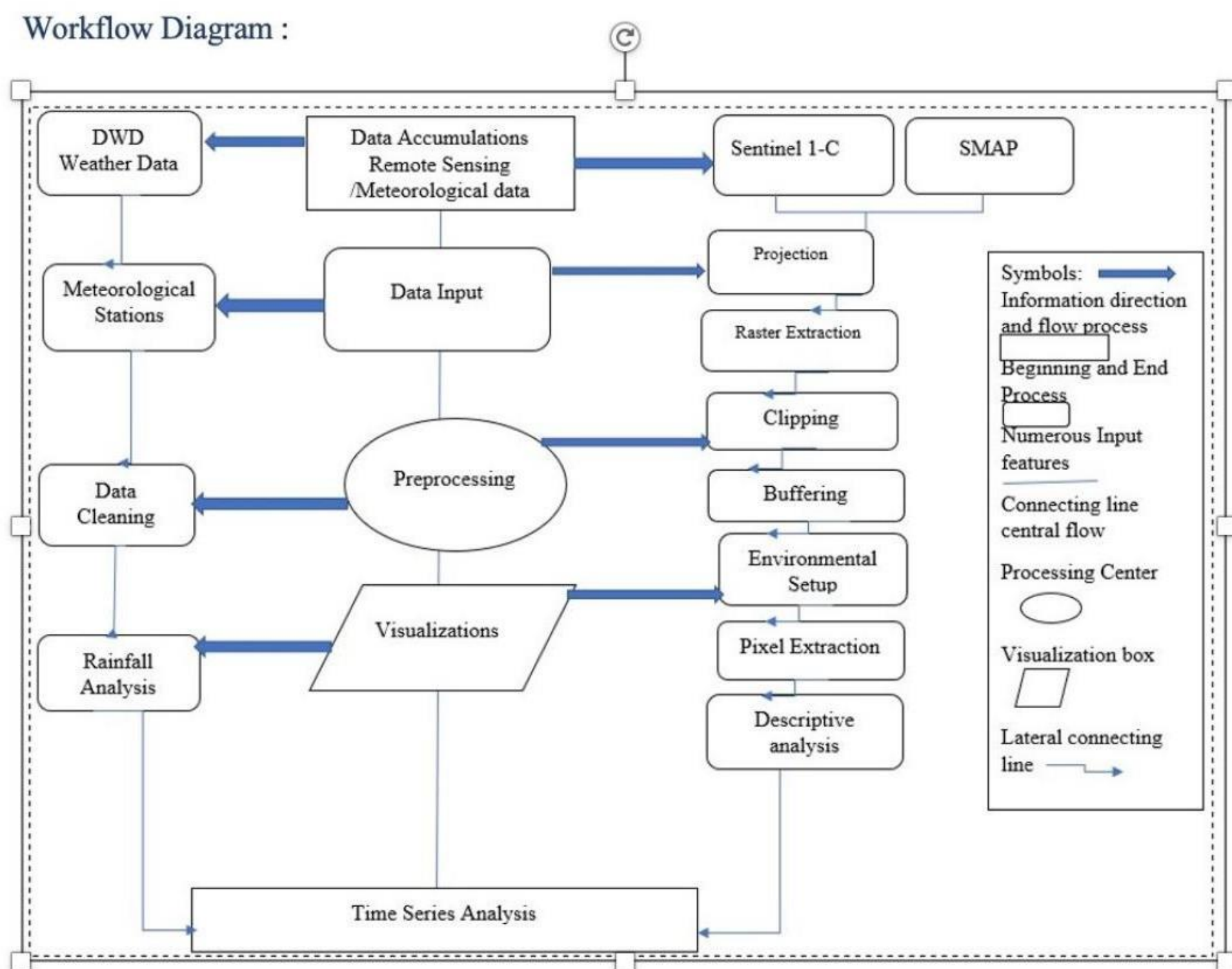
Satellite 2 — SMAP (Soil Moisture Active Passive)

9 km resolution · 5 cm surface soil moisture + 0–100 cm root-zone soil moisture

Precipitation — Deutscher Wetterdienst (DWD)

Stations near windthrow areas: 6 (Eberswalde/Brandenburg), 18 (NRW), 56 (Hessen). Used before and during the storm events (=12–13 weeks); satellite data tested pre- and post-storm (6 days).

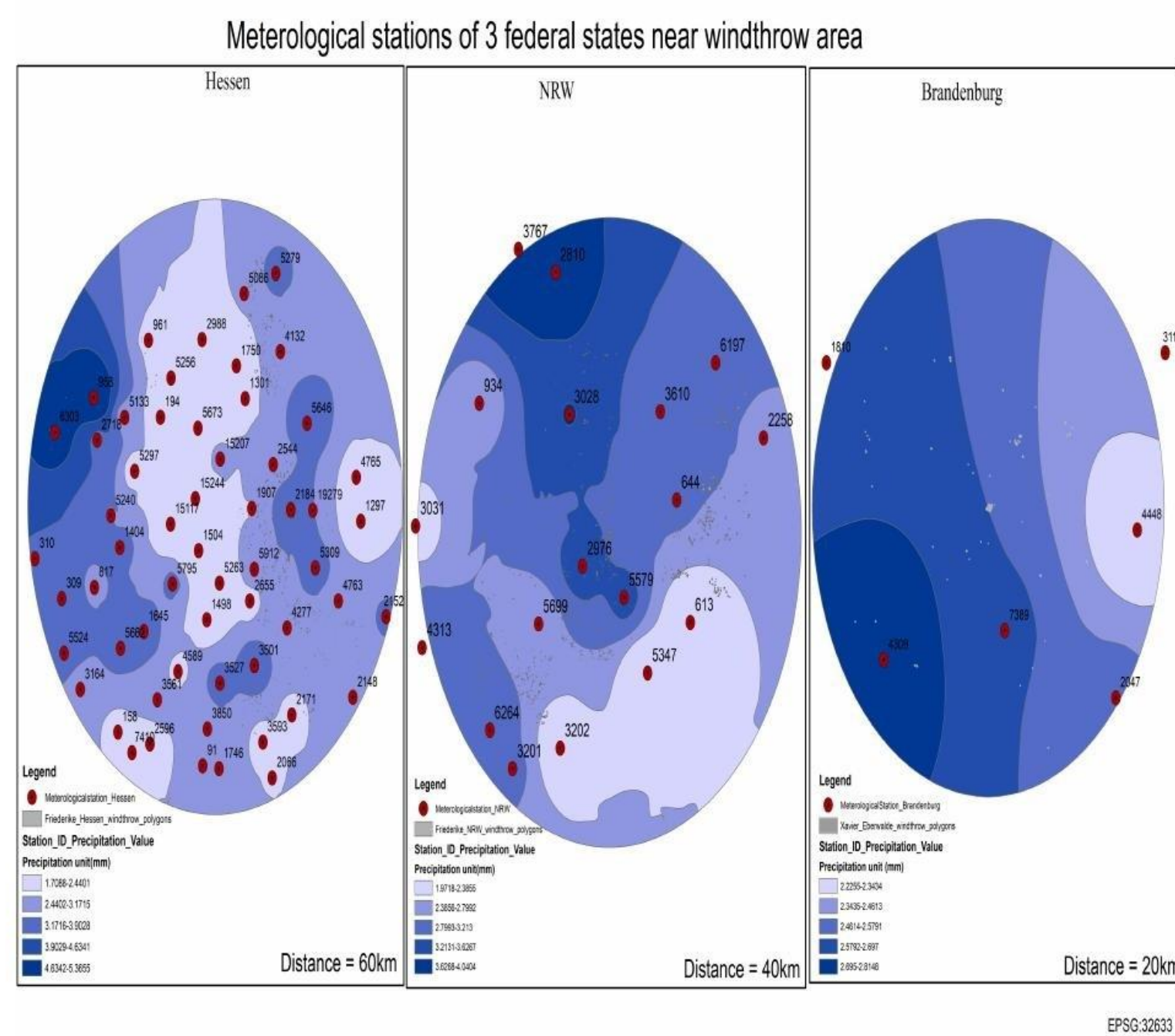
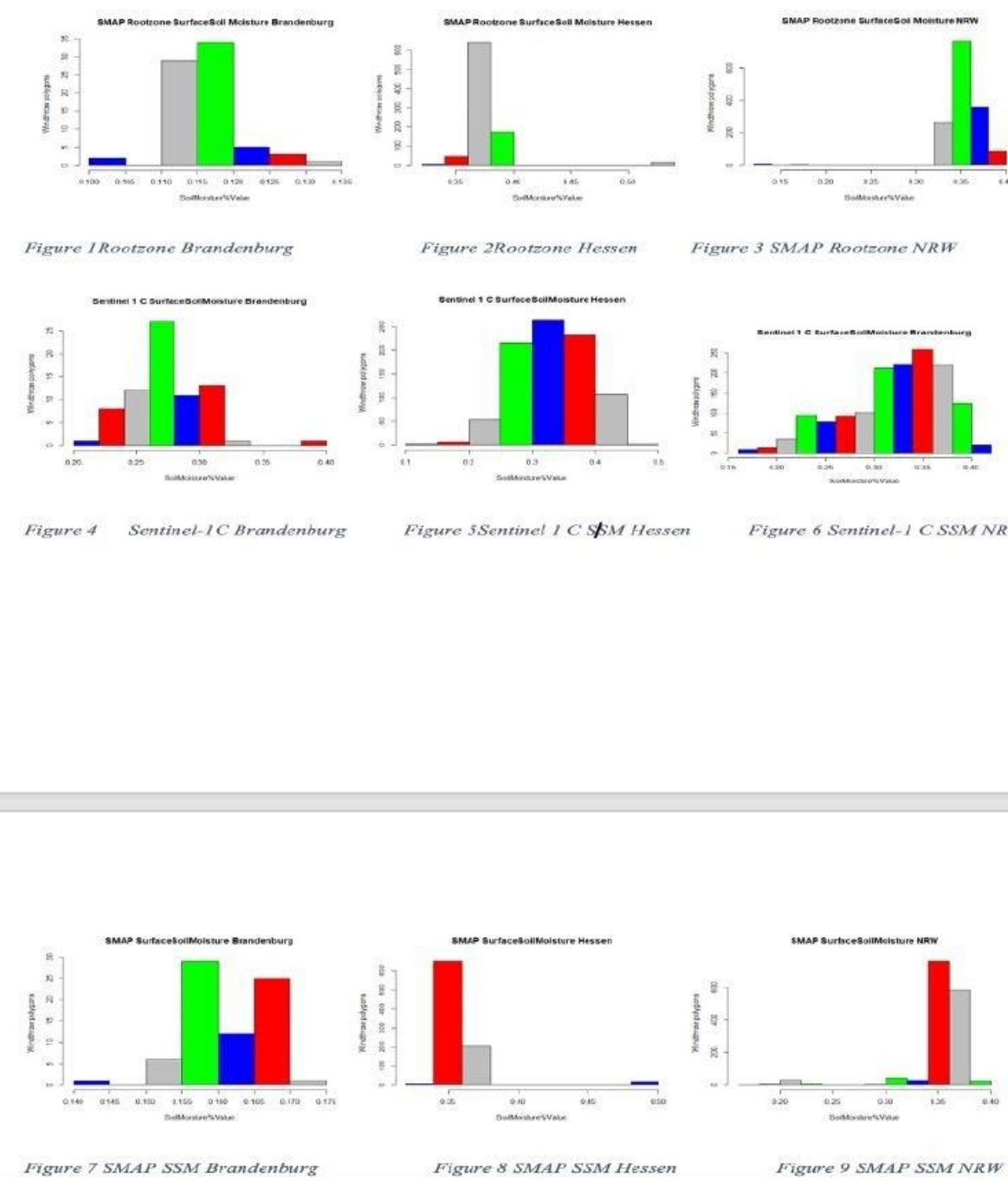
- Spatial analysis: clip, extract, overlay, count, calculate attributes, summary
- Buffering (1): zero / multiple-ring buffering around windthrow polygons
- Buffering (2): fixed distances – 50 m, 150 m, 1000 m
- Soil-moisture pixel extraction: (a) point sampling, (b) removing clusters
- Precipitation: locate weather stations per federal state; extract from web portal, clean, visualize



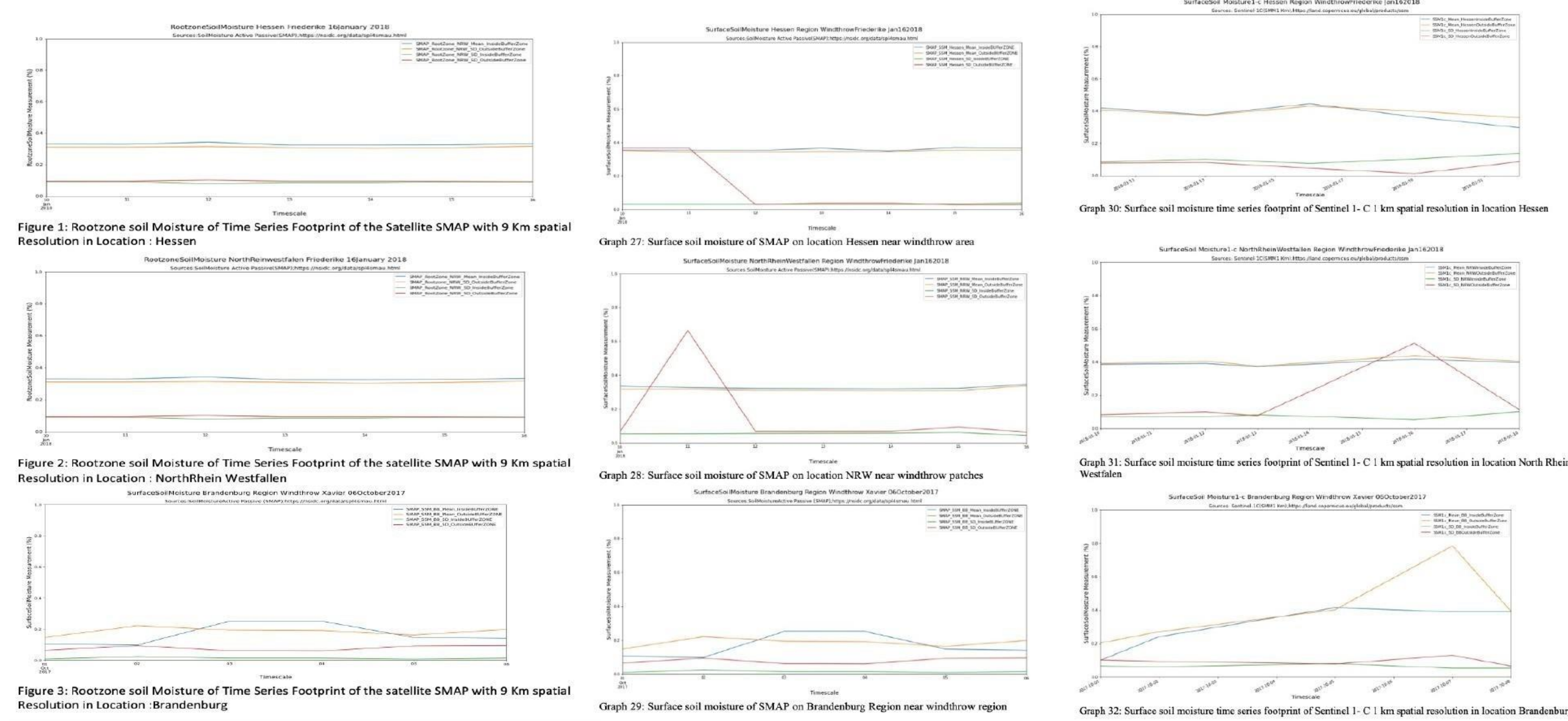
1. Both satellite products were cross-checked spatially and temporally against rainfall records at nearby weather stations; reasonable differences were noted between SMAP and 1 km Sentinel-1 surface soil moisture, and histogram analysis gives a more detailed picture of surface moisture than the Sentinel-1 distribution, which was uneven relative to precipitation events

Results (1) — Soil moisture on individual windthrow polygons & mean rainfall (rainfall interpolation, 3 states)

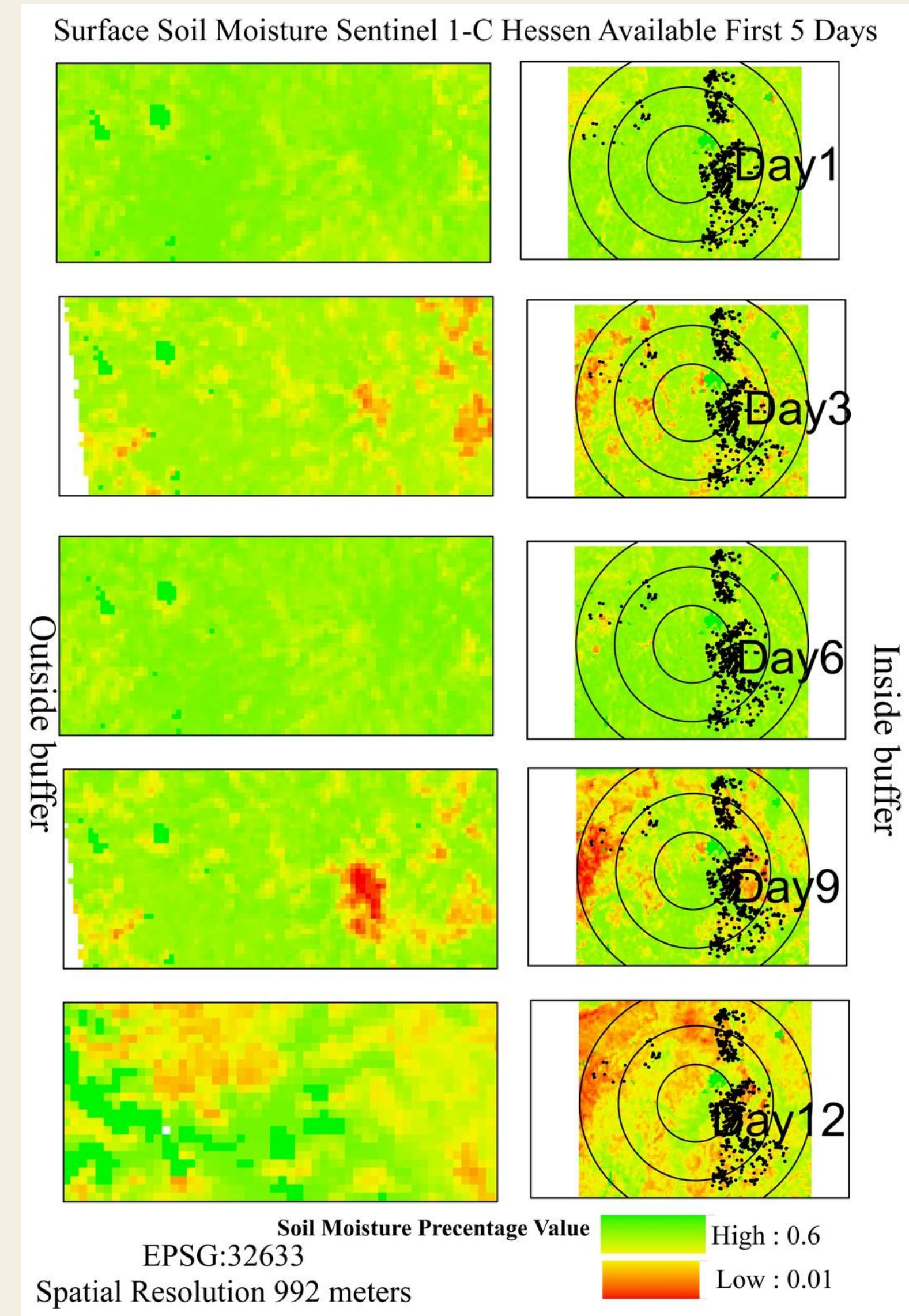
Soil moisture value of polygon



Root-zone & surface soil-moisture time series inside vs. outside the buffer zone (3 states,

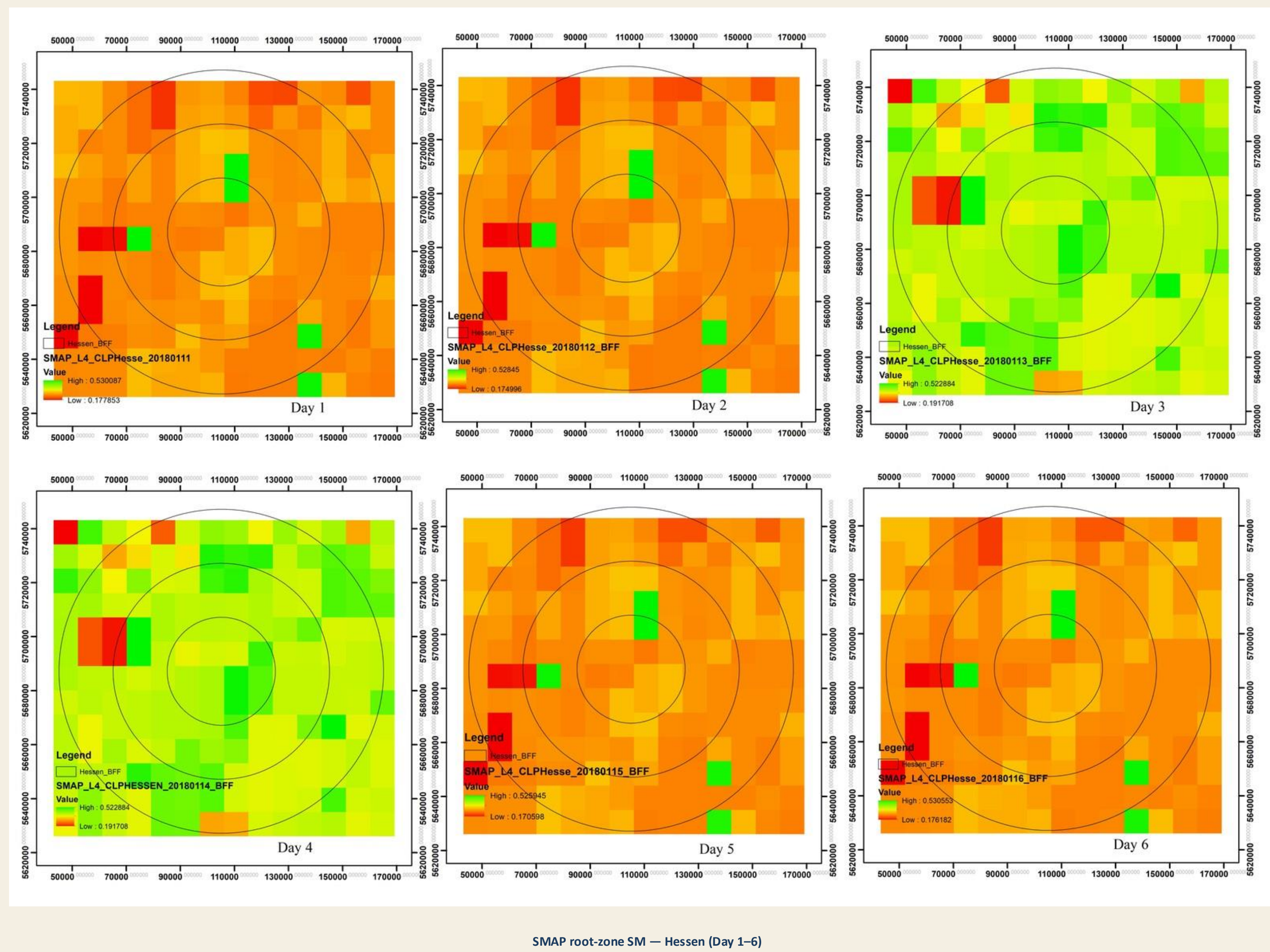


Daily satellite soil-moisture footprints inside vs. outside the buffer zone (higher-detail view)

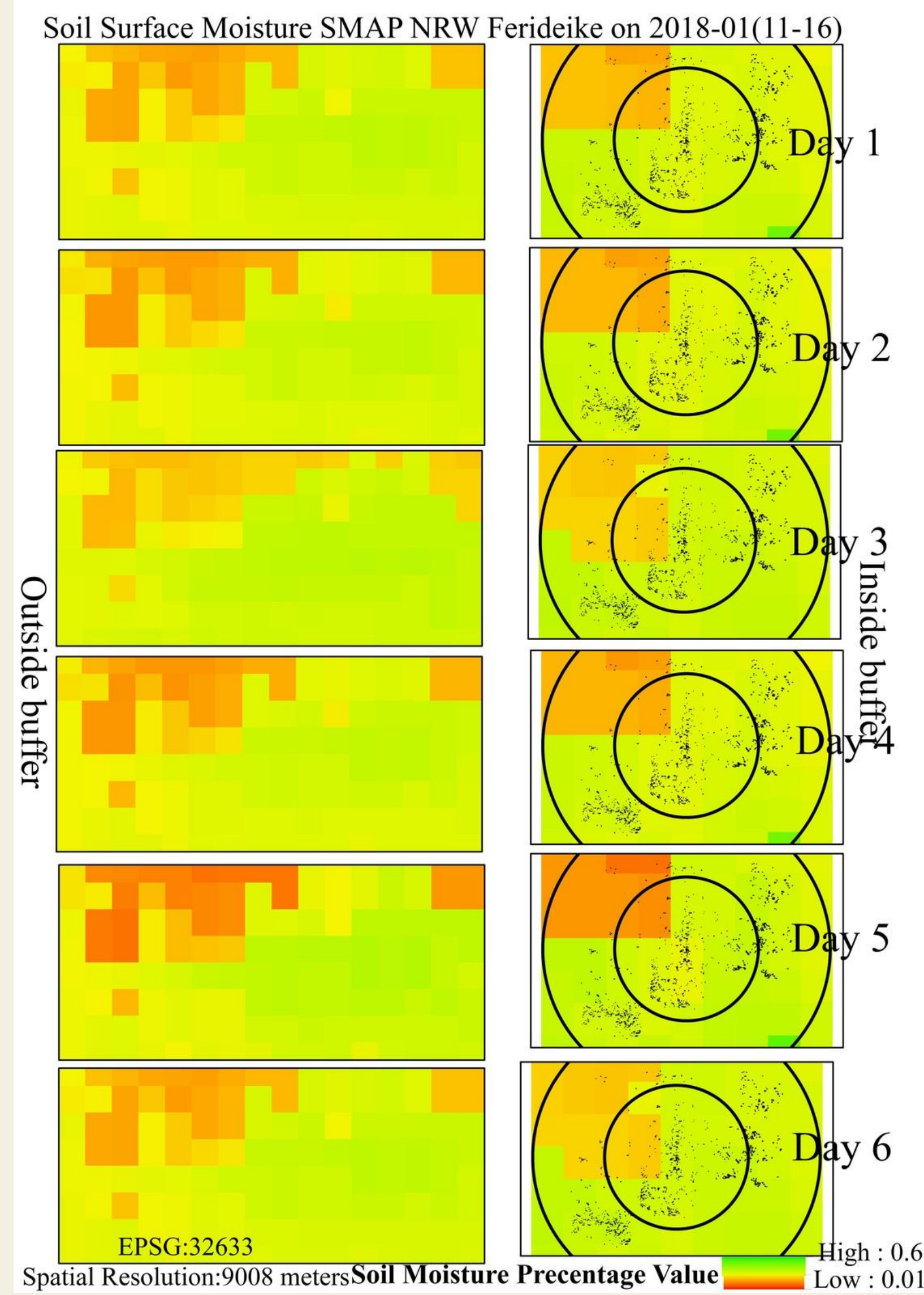


Sentinel-1 SSM — Hessen (Day 1,3,6,9,12)

Note: an equivalent SMAP surface soil-moisture series for Hessen, and the SMAP vs. Sentinel-1 footprint-coverage comparison, are available as supplementary figures.



SMAP root-zone SM — Hessen (Day 1–6)



EPSG:32633
 Spatial Resolution:9008 meters

Soil Moisture Percentage Value

High : 0.6
 Low : 0.01

SMAP surface SM — NRW (Day 1–6)